IMPERIAL

HRAC: History-Residual Adaptive Clipping for Byzantine-Robust Federated Learning

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Abstract

Federated learning enables collaborative training without centralising raw data, but it remains vulnerable to Byzantine clients and model poisoning. Robust aggregation is especially difficult under non-IID data, where benign clients may have persistent and legitimate update differences, and under temporal attacks that create slow cross-round drift.

This dissertation proposes *HRAC* (History-Residual Adaptive Clipping), an oracle-free aggregation method that performs bounded suppression rather than hard exclusion. HRAC combines an EMA-smoothed median norm cap with per-client history-residual clipping, then applies a delayed soft weighting rule based on residual-change statistics. By comparing clients primarily with their own update histories, HRAC aims to reduce harmful influence without forcing benign non-IID clients into a single global cluster.

Experiments on CIFAR-10 with 10 clients and 30% Byzantine participation show that HRAC remains competitive under benign training and improves robustness in several structured or temporally persistent attack settings, particularly non-IID MSA, HisMSA, ALIE, and IPM. Component ablations indicate that the global median cap is especially influential under MSA, while ν-based temporal weighting contributes most clearly under ALIE. The results support the use of lightweight client-specific residual histories, while also showing limitations under protocol-compliant label poisoning, bit flipping, and limited experimental coverage.

Declaration of Originality

I hereby declare that, unless otherwise indicated, the work presented in this thesis is entirely my own. To the best of my knowledge, it is original, and any ideas developed in collaboration with others have been appropriately acknowledged and referenced. I acknowledge the use of ChatGPT-5 (OpenAI, https://chatgpt.com) as a support tool to refine academic phrasing, develop the draft structure, and clarify concepts, and I confirm that all substantive content—including the literature review, analysis, designs, and conclusions—remains my own work. I further declare that I have not directly incorporated any AI-generated context in full into this submission.

Copyright Declaration

Contents

1 Introduction

Contents

1.1 Motivation

Machine learning systems are increasingly deployed across networks of devices that observe, decide, and interact with the physical world. In many deployments, training data is inherently distributed across mobile devices, embedded sensors, and edge gateways, and cannot be conveniently centralised due to privacy constraints, regulatory requirements, bandwidth limitations, or operational risk. Federated learning addresses this setting by enabling clients to compute model updates on local data and communicate only updates to a coordinating server.

However, decentralisation enlarges the threat surface. Standard aggregation rules, such as averaging, assume that received updates are broadly informative and aligned with the global learning objective. In realistic systems this assumption can be violated by device faults, software bugs, unreliable networks, or adversarial clients. Under such conditions, a small number of abnormal updates may dominate the aggregate, destabilise training, or steer the model towards unintended behaviour. Robust aggregation is therefore essential for trustworthy distributed learning.

Robustness is further complicated by client heterogeneity. Under non-identically distributed (non-IID) data, benign clients may legitimately differ in update direction and magnitude, and classical Byzantine-robust rules that rely on tight clustering can over-suppress these benign but diverse contributions. Finally, temporal dynamics introduce additional vulnerabilities: server-side momentum and other history-based mechanisms improve optimisation, but can be exploited by stealthy adversaries that apply small perturbations over many rounds, gradually biasing server state and inducing long-term drift. These challenges motivate history-aware and non-IID-tolerant aggregation methods that suppress malicious behaviour without relying on rigid thresholds or oracle knowledge of the attacker fraction.

1.2 Project Aims and Contributions

This project investigates robustness in federated learning under the Byzantine threat model, where a subset of clients may transmit arbitrary or adversarial updates. The aim is to design and evaluate a server-side aggregation mechanism that remains useful under both IID and non-IID data while resisting attacks that act through update magnitude, update structure, or persistent temporal drift. The central difficulty is that benign heterogeneity and malicious behaviour can look similar at the level of a single communication round: a non-IID client may legitimately differ from the population, while a Byzantine client may exploit that diversity to avoid simple outlier checks.

The main technical contribution is *HRAC* (History-Residual Adaptive Clipping), a history-aware aggregation rule that compares each client primarily with its own past behaviour rather than with a single global centre. HRAC combines a median-based global norm cap, per-client residual clipping, and soft weighting based on residual-change statistics. This design is intended to reduce the influence of harmful updates without turning the aggregator into a hard detector that requires attacker labels, an oracle attacker fraction, or a tight cluster of benign clients.

The project also contributes an implementation and evaluation pipeline for studying this design in controlled federated learning experiments. The pipeline supports interchangeable attacks and aggregation baselines, logs diagnostic quantities such as residual thresholds, μ/ν histories and client weights, and produces accuracy curves and ablation results for analysing robustness– utility trade-offs. These implementation and diagnostic components are part of the contribution because they make it possible to assess not only whether HRAC improves final accuracy in a given setting, but also which parts of the method are responsible for its behaviour.

The remainder of the report is organised as follows. Chapter 2 reviews federated learning, Byzantine-robust aggregation, poisoning attacks, and history-aware defences. Chapter 3 presents the HRAC method and its implementation. Chapter 4 evaluates HRAC against baseline aggregators and reports component ablations. The final chapters discuss professional issues, project management, limitations, and future work.

2

Background and Related Work

Contents

2.1 Federated Learning under Heterogeneity

Federated learning (FL) trains a shared model from decentralised data by iterating between local client optimisation and server-side aggregation. FedAvg and related model-averaging methods provide the standard starting point for this setting [1], [2]. Their appeal is practical: clients communicate model updates rather than raw data, reducing central data collection and allowing training across distributed devices. Large-scale FL systems add further constraints beyond the optimisation algorithm itself. Production deployments must handle partial participation, client availability, stragglers, communication limits, and privacy-preserving aggregation protocols [3], [4]. Benchmarking work such as LEAF also emphasises that realistic federated data are naturally heterogeneous across users and tasks [5].

The main complication for robust aggregation is that realistic FL is rarely IID. Client datasets can be label-skewed and unbalanced, and they are often shaped by local usage patterns [6]. As a result, benign client updates may legitimately differ in direction and magnitude. Optimisation work such as FedProx [7], SCAFFOLD [8], and adaptive server methods [9] address benign instability under heterogeneity, but these methods do not by themselves protect the server from malicious updates. Empirical studies of federated visual classification further show that the degree of non-IIDness can materially change FedAvg behaviour [10]. For this project, heterogeneity is not only an optimisation difficulty; it is a security difficulty, because benign diversity can be mistaken for adversarial behaviour. Recent Byzantine-robust FL work therefore treats heterogeneity as part of the security problem rather than as a secondary nuisance. Resampling and mixing strategies have been proposed to reduce the gap between benign heterogeneous clients before applying robust aggregation [11], [12]. HRAC follows the same broad motivation, but it avoids constructing mixed client batches and instead keeps a lightweight history for each individual client.

2.2 Byzantine-Robust Aggregation

Byzantine robustness studies aggregation when an unknown subset of workers may send arbitrary or coordinated messages [13], [14]. In distributed and federated learning, the aggregation rule becomes the main security boundary: the server typically observes only the submitted updates, not the clients' data or intentions. Robust aggregation methods therefore aim to form a useful global update even when some inputs are corrupted.

Existing defences can be grouped by their aggregation principle. Selection-based rules such as Krum choose updates that appear close to other updates [15]. Coordinate-wise methods such as median and trimmed mean apply robust statistics independently by coordinate [16]. High-dimensional robust estimation methods, including geometric-median-based RFA [17], estimate a central update with reduced sensitivity to outliers. Centered clipping instead limits each update relative to a centre and has been studied as a practical robust aggregation primitive [18]. Other lines of work directly address robustness under heterogeneous data, such as RSA, which incorporates a regularisation-based robust aggregation objective for non-IID distributed datasets [19]. Outside strictly Byzantine settings, adaptive clipping has also been used in private FL to tune clipping thresholds from update-norm statistics rather than from expensive manual search [20]. This line of work supports the use of norm-based, data-dependent caps, although its goal is privacy accounting rather than adversarial robustness.

These methods demonstrate that aggregation can be made more robust than simple averaging, but they also expose a recurring limitation: many defences rely, explicitly or implicitly, on a population-level notion of centrality. This is fragile when benign clients are heterogeneous. Selection-based rules can also hide vulnerabilities under coordinated attacks [21]. Variance-based attacks add a second difficulty: malicious updates can remain statistically plausible while still damaging training [22]. Recent robust-estimation work also notes a sample-efficiency trade-off: robust rules may need more participating clients than averaging to recover the same utility in clean settings [23]. These observations motivate robust methods that avoid treating all non-central clients as suspicious.

2.3 Model-Poisoning and Stealthy Attacks

Poisoning attacks can target the model update directly or corrupt local data so that protocol-following training produces harmful updates [6]. Model-poisoning attacks are often stronger because the attacker can craft the submitted update itself. Model replacement and backdoor attacks, for example, can scale malicious updates to dominate averaging while preserving benign-looking performance on clean data [24]. Follow-up work has studied when backdoors remain effective in realistic FL settings and how norm clipping changes the attack surface [25]. Other attacks explicitly degrade learning while evading robust aggregation rules [26]. Production-oriented evaluations further show that attack success depends strongly on deployment assumptions, threat-model realism, and the interaction between poisoning strategy and aggregation rule [27]. This makes it important to report not only final accuracy, but also the conditions under which a defence is expected to work.

Stealthy attacks are especially relevant to this project because they challenge simple outlier assumptions. ALIE exploits benign variance so that malicious updates remain within a statistically plausible range [22]. IPM and related sign or direction manipulations can push the aggregate away from the benign direction without requiring obviously large norms. Model Shuffle Attack (MSA) further manipulates parameter structure by shuffling locations and scaling updates, degrading differential-privacy-based FL while avoiding naive anomaly checks [28]. These attacks suggest that robustness cannot depend only on detecting extreme update magnitudes.

2.4 Data Poisoning and Label-Flipping Attacks

Data poisoning is weaker than arbitrary Byzantine update fabrication, but it is often more realistic. In label flipping, a malicious client changes part of its local labels and then follows the normal training protocol. The submitted update can therefore look plausible because it is produced by standard local optimisation rather than by direct message fabrication. Label-poisoning attacks and defences have been studied in FL settings [29], [30].

Recent work shows that this threat model can reverse common intuitions about robust aggregation. Gradient-space methods such as LFighter attempt to detect poisoned clients by clustering worker gradients [31], but clustering can become unreliable when benign data are strongly non-IID. More generally, under label poisoning and sufficient heterogeneity, mean aggregation can be theoretically and empirically more robust than several robust aggregators [32]. This result is important for HRAC: it argues against overly aggressive hard filtering and supports a soft suppression strategy that can preserve benign heterogeneous updates.

2.5 History-Aware Defences and Temporal Attacks

History is useful in distributed learning because it can stabilise noisy optimisation. Server-side momentum and adaptive optimisation use temporal state to improve convergence under stochasticity and partial participation [9]. Robust optimisation can also exploit history: learning-from-history frameworks construct robust directions from past information, and centered clipping with momentum demonstrates that historical state can improve Byzantine resilience [18]. Variance-reduced and momentum-based robust methods similarly argue that reducing stochastic variation can make malicious messages easier to separate from noisy benign updates [33], [34]. Historical-gradient studies also examine whether storing past client gradients helps detect attacks, but they highlight the storage and privacy costs of keeping high-dimensional historical update records [35]. This motivates compact temporal summaries rather than full-gradient histories.

The same temporal state can become an attack surface. A malicious client may apply small but persistent perturbations that are inconspicuous in a single round but accumulate over time. HisMSA explicitly models historical knowledge to craft poisoning updates that bypass multiple countermeasures [36]. This creates a design tension: a defence should use history, but it should do so in a way that detects persistent residual change rather than blindly trusting historical momentum.

2.6 Privacy, Deployability, and Lightweight Statistics

Practical FL systems may use privacy-enhancing mechanisms such as secure aggregation. In this setting, the server recovers only an aggregate sum rather than each individual update [37]. This constrains robust defences, since many anomaly detectors require detailed per-client visibility. Other approaches, such as FLTrust, use a server-side reference update to reweight client updates by alignment [38]. Such methods improve robustness but introduce additional assumptions about the server's trusted data or reference direction.

Scalability also matters. Pairwise distances, clustering, and repeated high-dimensional comparisons can become expensive as the number of clients and model dimension increase. Lightweight per-client statistics, such as update norms, residual scales, and residual-change summaries, offer a practical alternative. They can be maintained with exponential moving averages and used to detect abrupt or persistent behavioural changes without repeatedly solving a large robust-estimation problem. In decentralised learning, robustness is also affected by the communication topology rather than only by the aggregation formula; robust network topology studies show that system structure can change the resilience of distributed learning [39]. HRAC is narrower in scope because it uses a central server, but this literature reinforces the broader point that a deployable defence must account for system constraints.

2.7 Research Gap and Positioning

The related work points to four gaps that motivate HRAC. First, many robust aggregators compare clients to a population-level centre, which can suppress benign non-IID clients. Second, hard filtering can be brittle under structured data poisoning, where malicious updates are protocol-compliant and not necessarily extreme. Third, stealthy attacks can exploit benign variance or temporal state, making round-wise outlier detection insufficient. Fourth, robustness mechanisms must remain computationally lightweight enough to fit a federated learning pipeline.

HRAC is positioned as a response to these gaps. It uses a median-based global norm cap to limit extreme magnitudes, but it does not rely solely on a global cluster assumption. Instead, each client is compared against its own historical centre through residual clipping, and persistent residual movement is tracked with a lightweight ν statistic. The resulting penalty is soft rather than a hard exclusion rule, aiming to reduce malicious influence while retaining useful contributions from heterogeneous benign clients.

2.8 Conclusion

The literature motivates two design principles for this project. A robust FL aggregator should control harmful updates without assuming that all benign clients look similar, and it should use temporal information without becoming vulnerable to slow history-aware drift. HRAC follows these principles by combining EMA-smoothed median norm capping, per-client residual histories, and soft residual-change weighting.

3

Method and Implementation

Contents

3.1 Overview

This chapter defines HRAC and explains how it is implemented in the federated learning codebase. The design goal is to reduce malicious influence without forcing all benign clients into one tight cluster. This is important under non-IID data: a benign client may consistently differ from the population mean because its local data distribution is different. HRAC therefore uses each client's own history as the main reference. The method combines (i) a robust global norm cap, (ii) per-client history-residual clipping, (iii) residual-change tracking through EMA statistics, and (iv) a late-stage soft weighting rule based on elevated change statistics.

3.2 Problem Setup and Threat Model

Consider a federated learning system with N clients and a coordinating server. At communication round t, each participating client computes a local model update or gradient-like message $\Delta_i^t \in \mathbb{R}^d$ and sends it to the server. The server applies an aggregation rule

$$\mathrm{Agg}(\Delta_1^t, \ldots, \Delta_N^t)$$

to form a single update g_t, which is then used to update the global model.

The client population is divided into an unknown benign set $\mathcal{G}$ and a Byzantine set $\mathcal{B}$, with $|\mathcal{B}| = b$ and $|\mathcal{G}| = N - b$. Clients in $\mathcal{G}$ follow the training protocol, but their updates may be heterogeneous because their local data are non-IID. Clients in $\mathcal{B}$ may send arbitrary or strategically crafted messages. In the experiments, this includes both direct model manipulation attacks, such as bit flipping, ALIE, IPM, MSA, and HisMSA, and protocol-compliant data poisoning through label flipping.

The defence does not assume access to attacker labels, the benign aggregate, or the exact Byzantine identities during aggregation. The objective is therefore not to perfectly classify clients as benign or malicious. Instead, the goal is to limit the influence of harmful updates while preserving useful contributions from benign but heterogeneous clients. This distinction is important under non-IID data: a benign client may be far from the population centre for legitimate statistical reasons, so hard outlier removal can damage utility.

HRAC is designed for this setting with two assumptions. First, extreme round-wise update magnitudes should be bounded before aggregation. Second, client behaviour should be compared primarily with the client's own historical trajectory rather than only with a global population centre. The method remains oracle-free: labels identifying attackers are used only for controlled diagnostic analysis, not by the aggregation rule itself.

3.3 Aggregation Pipeline

In communication round t, the server receives flattened client updates $\{\Delta_i^t\}_{i=1}^N$, with $\Delta_i^t \in \mathbb{R}^d$. HRAC first limits extreme magnitudes using a median-based cap, then processes each client relative to its own residual history. After a warm-up period, the server applies a soft ν-based weight penalty before forming the aggregate and updating the per-client histories.

Algorithm 1 summarises the complete server-side HRAC procedure. The rest of this section gives the definitions and design rationale behind each stage.

3.4 Core Components

3.4.1 Global Median Norm Cap

The first defence layer prevents a single large update from dominating the aggregate. HRAC smooths the round-wise median update norm with an EMA before forming the clipping threshold. Let

$$m_t = \mathrm{med}_{1\le j\le N} \|\Delta_j^t\|_2, \qquad \bar{m}_t = \rho_g \bar{m}_{t-1} + (1-\rho_g) m_t.$$

The clipping threshold used in round t is formed from the stored scale at the start of the round,

$$B_t = c_g \bar{m}_{t-1},$$

with the first round warm-starting $\bar{m}$ from the current median norm. Thus, the current round contributes to the next round's cap rather than immediately changing its own threshold. This makes the cap predictable with respect to the current batch of submitted updates and reduces sensitivity to one-round norm spikes or drops. Each update is clipped by

$$\hat{\Delta}_i^t = \min\left(1, \frac{B_t}{\|\Delta_i^t\|_2 + \epsilon}\right) \Delta_i^t. \tag{3.1}$$

The implementation uses $c_g = 3.0$ and $\rho_g = 0.95$ by default. The cap remains median-based, so it does not require knowing which clients are malicious and is less sensitive to extreme Byzantine norms than a mean-based scale estimate; the additional EMA smoothing makes the cap less volatile across communication rounds.

3.4.2 Per-Client History-Residual Clipping

For each client i, HRAC stores a historical centre b_i^t. This centre is an EMA of safe past messages and represents the client's own typical update path, not a population-wide prototype. The residual

Algorithm 1 History-Residual Adaptive Clipping (HRAC)

Require: updates $\{\Delta_i^t\}_{i=1}^N$, client ids, states $(b_i, \mu_i, \nu_i, \bar{r}_i^{\text{prev}}, \bar{m})$, parameters $c_g, c, \rho_g, \rho_b, \rho_\mu, \rho_\nu, T_\nu, \alpha, w_{\max}$
Ensure: aggregated update g_t

$m_t \leftarrow \text{med}_j \, \|\Delta_j^t\|_2$
if $\bar{m}$ is uninitialised, set $\bar{m} \leftarrow m_t$; $\quad B_t \leftarrow c_g \bar{m}$
for each client i **do**
  $\hat{\Delta}_i^t \leftarrow \min(1, B_t/(\|\Delta_i^t\|_2 + \epsilon))\Delta_i^t$
  if i is a new client **then**
    initialise $b_i, \mu_i, \nu_i, \bar{r}_i^{\text{prev}}$
    $\tilde{\Delta}_i^t \leftarrow \hat{\Delta}_i^t$
  else
    $r_i^t \leftarrow \hat{\Delta}_i^t - b_i$; $\quad \tau_i^t \leftarrow c\mu_i + \epsilon$
    $\bar{r}_j^t \leftarrow \min(1, \tau_i^t/(\|r_i^t\|_2 + \epsilon))r_i^t$
    $\tilde{\Delta}_i^t \leftarrow b_i + \bar{r}_i^t$
    optionally cap $\tilde{\Delta}_i^t$ again by B_t
  end if
end for
if $t \le T_\nu$ **then**
  $w_i^t \leftarrow 1/N$ for all clients
else
  $\bar{\nu}^t \leftarrow N^{-1} \sum_{j=1}^N \nu_j$
  for each client i **do**
    $e_i^t \leftarrow \mathbb{I}[\nu_i > 0.9] \max(\nu_i - \bar{\nu}^t, 0)$
    $\hat{w}_i^t \leftarrow 1/(1 + \alpha e_i^t)$
  end for
  normalise $\{\hat{w}_i^t\}$ and cap weights by $w_{\max}$
end if
$g_t \leftarrow \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t$
for each non-new client i **do**
  $b_i \leftarrow \rho_b b_i + (1 - \rho_b)\tilde{\Delta}_{i,\text{safe}}^t$
  $\mu_i \leftarrow \rho_\mu \mu_i + (1 - \rho_\mu)\|\bar{r}_i^t\|_2$
  $d_i^t \leftarrow \|\bar{r}_i^t - \bar{r}_i^{\text{prev}}\|_2$
  $\nu_i \leftarrow \rho_\nu \nu_i + (1 - \rho_\nu)d_i^t$; $\quad \bar{r}_i^{\text{prev}} \leftarrow \bar{r}_i^t$
end for
$\bar{m} \leftarrow \rho_g \bar{m} + (1 - \rho_g)m_t$
return g_t

is

$$r_i^t = \hat{\Delta}_i^t - b_i^t. \tag{3.2}$$

The residual threshold is adaptive:

$$\tau_i^t = c\mu_i^t + \epsilon, \tag{3.3}$$

where μ_i^t is an EMA scale statistic and $c = 2.5$ by default. The clipped residual is

$$\bar{r}_i^t = \min\left(1, \frac{\tau_i^t}{\|r_i^t\|_2 + \epsilon}\right) r_i^t. \tag{3.4}$$

The processed message is then reconstructed as

$$\tilde{\Delta}_i^t = b_i^t + \bar{r}_i^t. \tag{3.5}$$

3.4.3 Separated Aggregation and Statistics Paths

The implementation distinguishes the aggregation path from the statistics path. Aggregation uses the globally clipped update $\hat{\Delta}_i^t$ and its processed version $\tilde{\Delta}_i^t$. The statistics path can lift extremely small-norm updates to the median norm when computing μ_i and ν_i. This prevents a near-zero update from artificially collapsing a client's history statistics. The aggregation value itself is not lifted in this case, so the actual model update remains faithful to the received message after global clipping.

3.4.4 EMA History Updates

After aggregation, HRAC updates the client histories. The aggregation-path centre is

$$b_i^{t+1} = \rho_b b_i^t + (1 - \rho_b)\Delta_{i,\text{safe}}^t, \tag{3.6}$$

where $\Delta_{i,\text{safe}}^t$ is the processed message after the safety cap. The default value is $\rho_b = 0.98$.

The residual scale statistic is updated as

$$\mu_i^{t+1} = \rho_\mu \mu_i^t + (1 - \rho_\mu)\|\bar{r}_i^t\|_2, \tag{3.7}$$

with a small floor $\mu_{\min}$ for numerical stability. The default is $\rho_\mu = 0.95$.

The change statistic ν_i is updated from the movement of the clipped residual:

$$d_i^t = \|\bar{r}_i^t - \bar{r}_i^{t-1}\|_2, \qquad \nu_i^{t+1} = \rho_\nu \nu_i^t + (1-\rho_\nu) d_i^t, \tag{3.8}$$

again with a small floor $\nu_{\min}$. The class default is $\rho_\nu = 0.95$. In the reported experiment driver, this value is set to $\rho_\nu = 0.87$ so that the residual-change statistic reacts slightly faster to persistent movement.

3.4.5 Soft Weighting from the ν Statistic

The weighting rule is deliberately delayed. HRAC does not immediately down-weight clients from the first round. For iterations $t \le T_\nu$, it uses uniform weights. In the implementation, $T_\nu = 50$ by default. This warm-up allows the per-client statistics to become meaningful before they affect aggregation.

Once the penalty is active, the server forms a population baseline

$$\bar{\nu}^t = \frac{1}{N} \sum_{j=1}^{N} \nu_j^t. \tag{3.9}$$

Only clients with sufficiently high ν_i^t are penalised. In code, the activation threshold is 0.9, and the excess score is

$$e_i^t = \begin{cases} \max(\nu_i^t - \bar{\nu}^t, 0), & \nu_i^t > 0.9, \\ 0, & \nu_i^t \le 0.9. \end{cases} \tag{3.10}$$

The raw weight is

$$\hat{w}_i^t = \frac{1}{1 + \alpha e_i^t}, \tag{3.11}$$

where $\alpha = 5.0$ by default. Raw weights are normalised to sum to one. A maximum per-client weight cap, $w_{\max} = 0.30$, is then applied iteratively with renormalisation to avoid a small number of clients dominating the round.

The final aggregate is

$$g_t = \sum_{i=1}^{N} w_i^t \tilde{\Delta}_i^t. \tag{3.12}$$

3.5 Software Integration and Logging

HRAC is implemented as the `C_HRAC` aggregation class in the same centralised aggregation interface used by the baseline rules. This design keeps the training loop independent of the chosen aggregator: the server receives a list of client messages, calls the selected aggregation object, and applies the returned update to the global model. In the experiment driver, selecting `hrac` instantiates `C_HRAC` with the same client sets and attack configuration used by the other aggregators, which allows HRAC to be compared against mean aggregation, FABA, centred clipping, and LFighter under matched training settings.

The attack layer is also modular. The same training entry point can construct label-flipping attacks, bit flipping, ALIE, IPM, MSA, or HisMSA before passing the attack object to the centralised training environment. This separation is important for the evaluation: HRAC is not written as a special case for one attack, and the aggregation rule does not receive attacker labels during training. Attacker identities are supplied only to the controlled logging code so that diagnostic summaries can report, for example, whether low-weight clients correspond to known Byzantine clients in a simulated run.

The implementation exposes component switches for ablation experiments. The driver can disable the global median cap, disable per-client residual clipping, disable ν-based weighting, or leave only the global cap before averaging. These switches reuse the same `C_HRAC` implementation, changing only the enabled components. This avoids maintaining separate aggregator classes for each ablation and reduces the risk that observed differences are caused by unintended changes elsewhere in the training pipeline.

In addition to returning the aggregate update, HRAC records diagnostic quantities at a fixed logging interval. The log contains residual thresholds, μ and ν statistics, pre- and post-clipping norms, normalised weights, and attacker-aware summaries when attacker identities are known for evaluation. The implementation also includes invariant checks for numerical sanity, such as weight normalisation, non-negative weights, norm-cap constraints, and EMA update consistency. Training records are written through the common cache interface, which stores accuracy and loss trajectories together with experiment metadata such as dataset, partition, Byzantine level, seed, and ablation flags. The same cached records are then used by the plotting scripts that generate the figures reported in Chapter 4.

3.6 Hyperparameters

The main HRAC hyperparameters are:

- $\rho_b = 0.98$: EMA decay for the client history centre.
- $\rho_\mu = 0.95$: EMA decay for the residual scale statistic.
- $\rho_\nu = 0.87$ in the experiment driver: EMA decay for the residual-change statistic. The class default is 0.95, but the reported runs use a slightly faster ν update.
- $c = 2.5$: multiplier for the per-client residual clipping threshold.
- $c_g = 3.0$: multiplier for the EMA-smoothed global median norm cap.
- $\rho_g = 0.95$: EMA decay for the global median-norm scale.
- $T_\nu = 50$: first iteration after which ν-based weighting is used.
- $\alpha = 5.0$: strength of the ν excess penalty.
- $w_{\max} = 0.30$: maximum normalised weight allowed for one client.

The algorithm is intentionally not a hard detector. Client identities are used only for controlled logging and evaluation; the aggregation rule itself does not need attacker labels or the attacker fraction. The most sensitive parameters are c, which controls how much residual deviation is allowed before clipping, and α, which controls how sharply elevated ν values reduce weights.

3.7 Convergence Statement

The empirical evaluation in Chapter 4 is the primary evidence for HRAC's practical robustness, but the method also admits a modular convergence interpretation. The analysis treats HRAC as an aggregation rule that produces g_t, compares this aggregate with the ideal honest-client average, and then inserts the resulting aggregation error into the standard descent argument for smooth non-convex optimisation.

Under bounded honest updates, an honest-majority condition for the median-based scale, bounded honest clipping distortion, and a bounded residual Byzantine weight mass, HRAC converges to a neighbourhood of first-order stationarity. In particular, for a sufficiently small constant step size η, the averaged squared gradient norm satisfies a bound of the form

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(x_t)\|^2 \leq \frac{2(f(x_0)-f^\star)}{\eta T} + R_{\mathrm{HRAC}}, \tag{3.13}$$

where R_{HRAC} collects the honest stochastic variance, non-IID/bias term, global clipping distortion, residual clipping distortion, and the remaining Byzantine aggregation error. The EMA-smoothed median cap is handled by showing that the median scale remains bounded under an honest majority and that the EMA update preserves this bound.

This result should be read as a stability guarantee rather than as a claim of exact convergence to a zero-gradient point under arbitrary attacks. Persistent heterogeneity and nonzero Byzantine mass create a nonzero error floor. The full proof, including the aggregation-error decomposition and the explicit neighbourhood constant, is given in Appendix A.

4
Experiments and Results

Contents

4.1 Experimental Setup

The experiments are intended to evaluate both benign utility and Byzantine robustness under the same implementation pipeline. Table 4.1 summarises the common settings used for the CIFAR-10 results. The baseline condition uses $b = 0$ to verify that HRAC does not substantially harm normal training. Attack conditions use $b = 3$ out of $N = 10$ clients, corresponding to 30% malicious participation.

Table 4.2 makes the experimental scope explicit. The reported curves should be interpreted as controlled single-seed trends rather than as a full statistical robustness benchmark. This choice keeps the comparison focused on attack-dependent behaviour and component effects, but it also means that sensitivity to seeds, larger client populations, and different Byzantine fractions remains future work.

Table 4.1: Experimental setup for the CIFAR-10 evaluation.

Item	**Setting**
Dataset	CIFAR-10
Client population	$N = 10$ clients
Byzantine level	$b = 3$ malicious clients for attack settings; $b = 0$ for benign baseline
Data partitions	IID and label-separation non-IID partitions
Training horizon	20,000 communication iterations in the logged curves
Attack settings	Label flipping, MSA, HisMSA, bit flipping, ALIE, and IPM
Aggregation baselines	Mean, FABA, centred clipping (CC), LFighter, and HRAC variants where available
Main metric	Test accuracy over communication iterations
Diagnostics	Residual thresholds, μ/ν statistics, normalised weights, and attacker-aware summaries in controlled runs

Table 4.2: Coverage and scope of the reported experiments.

Dimension	**Covered in this report**	**Not covered systematically**
Dataset	CIFAR-10	Additional datasets and model families
Client population	$N = 10$ clients	Larger or varying client populations
Byzantine fraction	$b = 3$ for attacks, i.e. 30%; $b = 0$ benign baseline	Multiple attacker fractions
Data heterogeneity	IID and label-separation non-IID partitions	Multiple non-IID severity levels
Randomness	Fixed cached runs used consistently across methods	Multi-seed confidence intervals
Hyperparameters	One fixed HRAC configuration across main runs	Systematic hyperparameter sweeps

All methods are compared using the same cached training logs and plotting pipeline. The figures report accuracy trajectories rather than only the final round because several attacks differ in when they begin to degrade training: some cause early instability, while history-aware attacks can produce slower cross-round drift. For this reason, both trajectory shape and final performance are considered when interpreting robustness.

4.2 Experimental Results

Figures 4.1–4.3 summarise the main CIFAR-10 experiments used in this report. The plots compare HRAC against mean aggregation and robust baselines under IID and non-IID client partitions. The training setup uses 10 clients; unless otherwise stated, the Byzantine level is $b = 3$, corresponding to 30% malicious participants. The horizontal axis reports communication iterations, and the vertical axis reports test accuracy. The non-IID partition is the label-separation setting used by the implementation logs.

Figure 4.1 first checks benign utility and a protocol-compliant poisoning setting. Under benign training ($b = 0$), HRAC tracks the strongest baselines closely, indicating that the proposed aggregation rule does not introduce a large utility penalty when all clients are honest. Under label flipping, HRAC improves the early IID trajectory and remains competitive, but FABA and mean aggregation can be close or better in some regimes, especially under non-IID data. This suggests that protocol-compliant data poisoning is a different failure mode from direct model manipulation: poisoned clients can still produce updates that look plausible under the normal training protocol.

Figure 4.2 shows attacks that manipulate update structure or exploit temporal state. In the MSA panels, mean aggregation degrades severely, especially under non-IID data, whereas HRAC maintains a much higher trajectory. In the HisMSA panels, HRAC remains close to the best curves and is particularly stable in the non-IID setting, which is consistent with the design motivation for using client-specific history. These results show that HRAC is most effective when the attack creates structured or persistent deviation rather than only noisy benign variation.

Figure 4.3 reports bit flipping, ALIE, and IPM. The IID results are relatively close across several methods, but the non-IID panels separate the defences more clearly. HRAC is consistently among the strongest and most stable methods for non-IID ALIE and IPM, while bit flipping remains a more mixed case where FABA can perform slightly better. Overall, the results support the central design claim of HRAC: comparing clients against their own histories can improve robustness to structured and history-aware attacks while preserving benign training behaviour. At the same time, the mixed label-flipping and bit-flipping panels show that HRAC should be treated as a robust aggregation strategy rather than a universally dominant detector.

4.3 Additional Analyses

Beyond the main comparison curves, the implementation was also used to isolate which parts of HRAC are responsible for the observed behaviour. The component study below is reported for non-IID CIFAR-10, where benign heterogeneity makes hard filtering risky and where the main robustness claims are most relevant. The remaining analysis focuses on what can be concluded from the available logs, while broader sensitivity studies are treated as limitations and future work.

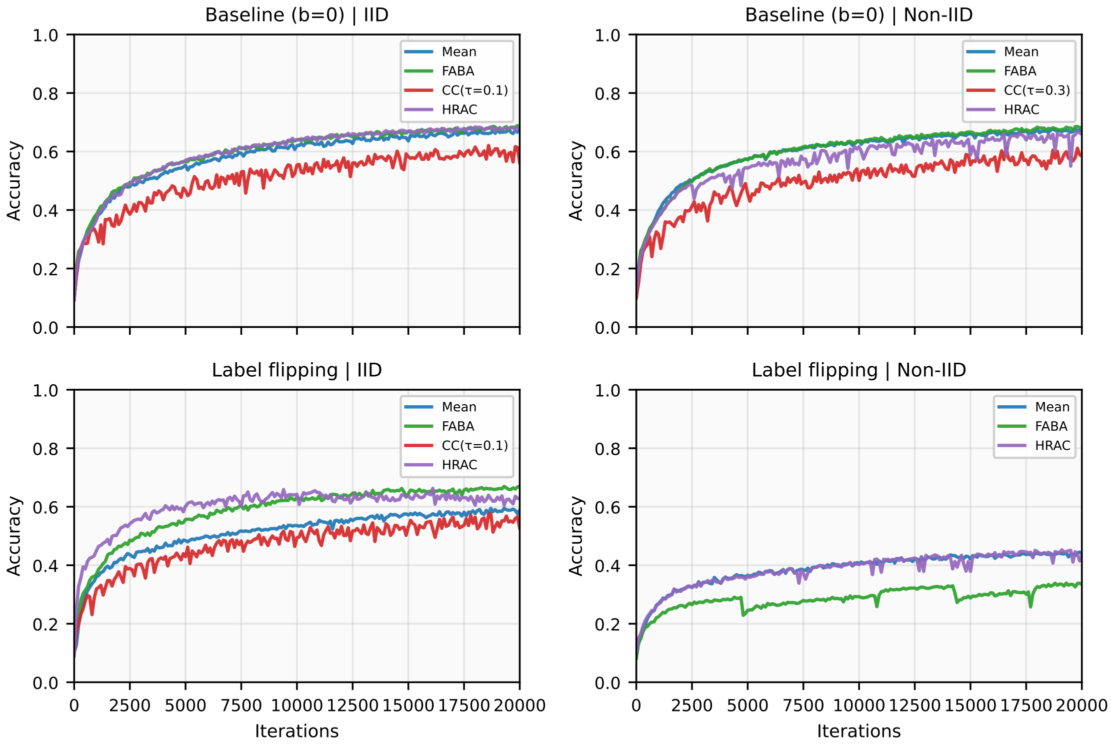

Figure 4.1: CIFAR-10 accuracy under benign training and label-flipping poisoning. The top row uses $b = 0$; the label-flipping row uses $b = 3$.

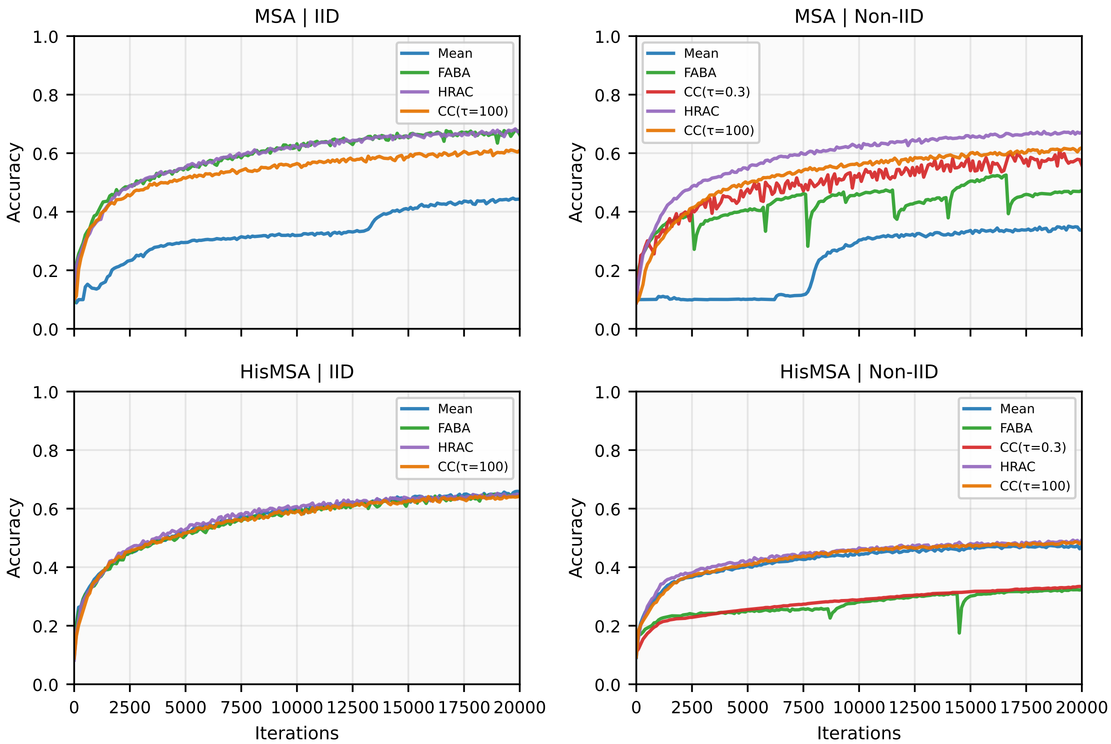

Figure 4.2: CIFAR-10 accuracy under model-shuffling (MSA) and history-aware model-shuffling (HisMSA) attacks with $b = 3$.

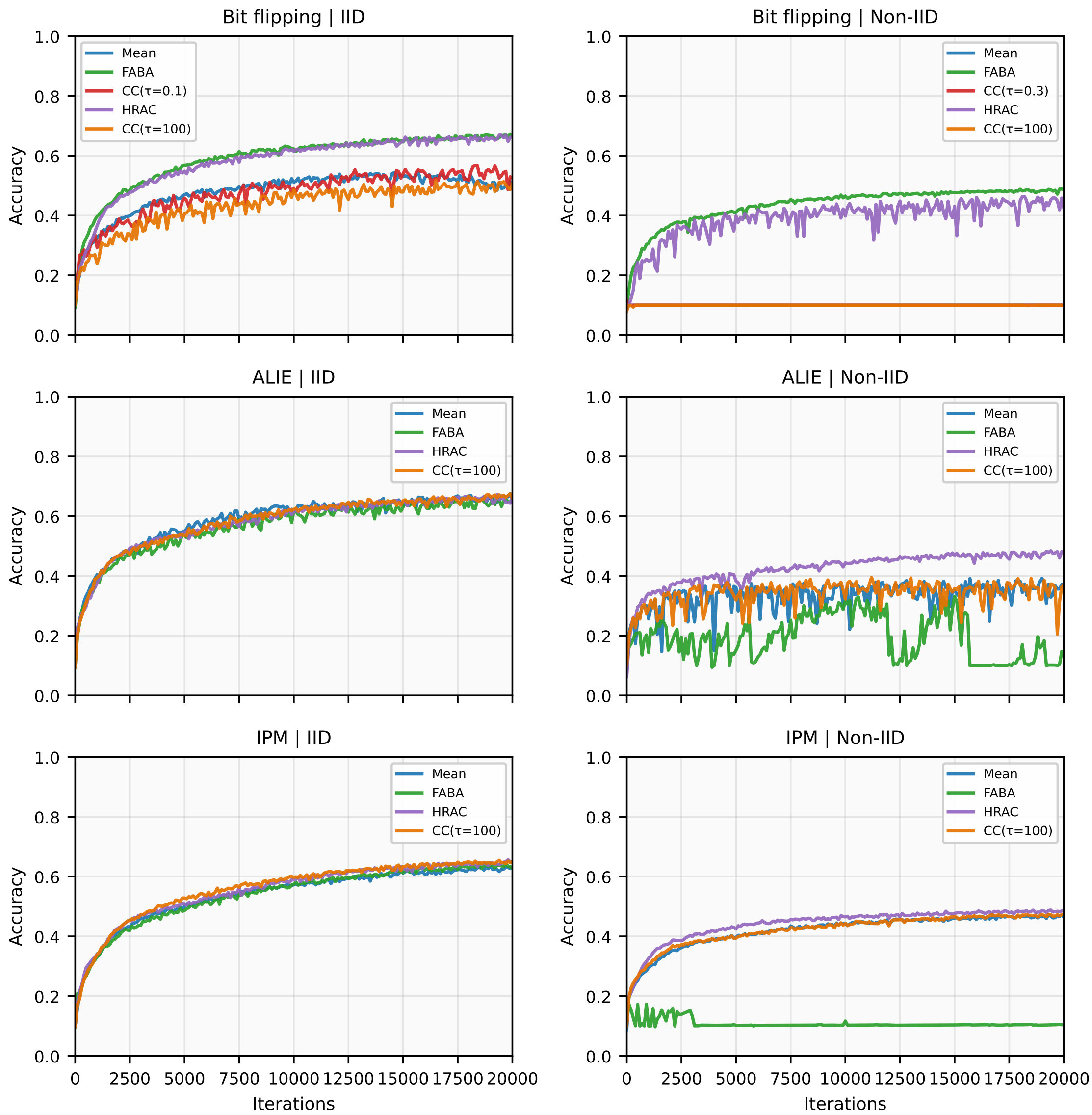

Figure 4.3: CIFAR-10 accuracy under bit-flipping, ALIE, and IPM attacks with $b = 3$.

Table 4.3: Final and best CIFAR-10 test accuracy extracted from cached logs. Entries are reported as final / best.

Setting	Partition	Mean	Best robust baseline	HRAC
Baseline	IID	0.6707 / 0.6780	FABA: 0.6808 / 0.6887	0.6808 / 0.6845
Baseline	Non-IID	0.6718 / 0.6770	FABA: 0.6820 / 0.6857	0.6590 / 0.6687
Label flipping	IID	0.5889 / 0.5951	FABA: 0.6697 / 0.6697	0.6271 / 0.6629
Label flipping	Non-IID	0.4391 / 0.4449	FABA: 0.3348 / 0.3393	0.4324 / 0.4527
MSA	IID	0.4433 / 0.4464	FABA: 0.6630 / 0.6789	0.6745 / 0.6832
MSA	Non-IID	0.3369 / 0.3507	CC($\tau = 100$): 0.6189 / 0.6189	0.6713 / 0.6743
HisMSA	IID	0.6577 / 0.6593	FABA: 0.6478 / 0.6500	0.6433 / 0.6523
HisMSA	Non-IID	0.4734 / 0.4743	CC($\tau = 100$): 0.4863 / 0.4868	0.4934 / 0.4934
ALIE	IID	0.6497 / 0.6758	CC($\tau = 100$): 0.6607 / 0.6753	0.6665 / 0.6722
ALIE	Non-IID	0.3585 / 0.3921	CC($\tau = 100$): 0.3695 / 0.3953	0.4799 / 0.4843
IPM	IID	0.6315 / 0.6348	CC($\tau = 100$): 0.6512 / 0.6546	0.6505 / 0.6562
IPM	Non-IID	0.4646 / 0.4733	CC($\tau = 100$): 0.4736 / 0.4740	0.4879 / 0.4880

4.3.1 Final and Best Accuracy Summary

Table 4.3 complements the accuracy curves by reporting final-round and best observed test accuracy. Each entry is written as “final / best”. The best robust baseline column excludes both mean aggregation and HRAC, so that mean aggregation remains visible as a separate reference.

4.3.2 Component Ablation Study

The ablation study considers four modifications of HRAC. Removing the global cap disables the EMA-smoothed median norm bound. Removing residual clipping keeps the global cap but no longer clips deviations from the per-client history. Removing ν weighting keeps the clipping stages but uses uniform aggregation weights throughout training. The “global cap only” variant removes both history-residual clipping and ν weighting, leaving only the smoothed median norm cap before averaging. These variants test whether HRAC’s robustness comes from a single clipping heuristic or from the combination of magnitude control, client-specific history, and temporal weighting.

Figures 4.4 and 4.5 should be read together because they expose different failure modes. Under ALIE, removing either the global cap or the history-residual clipping stage gives a trajectory close to full HRAC, and in this single run the final accuracy is slightly higher than full HRAC (Table 4.4). This is consistent with the nature of ALIE: the malicious updates are designed to remain statistically plausible by exploiting benign variance rather than by sending obviously large vectors. Therefore, pure magnitude control and residual clipping are not the only decisive mechanisms in this setting. The large degradation when ν weighting is removed, and the weaker global-cap-only result, indicate that the temporal residual-change statistic is the main source of robustness for this ALIE experiment.

The MSA ablation shows the opposite pattern. Under MSA, removing the global cap collapses

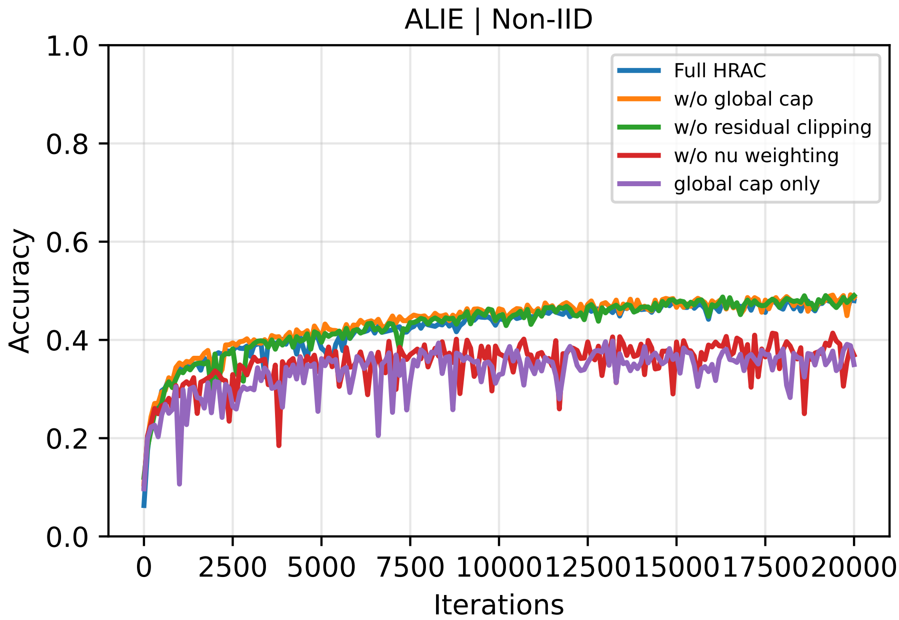

Figure 4.4: HRAC component ablation under ALIE on non-IID CIFAR-10 with $b = 3$.

Table 4.4: Final and best test accuracy for the ALIE non-IID ablation study.

Variant	**Final**	**Best**	**Main effect**
Full HRAC	0.4799	0.4843	Reference method
No global cap	0.4863	0.4919	Similar in ALIE
No residual clipping	0.4903	0.4903	Similar in ALIE
No ν weighting	0.3691	0.4139	Large degradation
Global cap only	0.3498	0.3979	Large degradation

final accuracy to approximately chance level, and removing residual clipping also introduces a visible performance gap relative to full HRAC (Table 4.5). This is because MSA can send structurally manipulated malicious updates with larger effective scales or layer-wise distortions. Without the global median cap, these updates can still enter the server aggregation path with enough magnitude to steer the global model update. Without residual clipping, the server still has the initial norm cap, but it no longer constrains each client relative to its own historical centre, so structured deviations can more easily affect the model trajectory and the client history state.

Together, these two ablations show that HRAC's components are threat-dependent rather than uniformly important in every setting. For variance-bounded attacks such as ALIE, full HRAC behaves similarly to the versions without the global cap or without residual clipping, while ν-based weighting is the critical part. For structural attacks such as MSA, the global median cap is essential and residual clipping provides an additional history-aware constraint. This supports the combined HRAC design: the method needs temporal weighting for stealthy plausible updates, and it needs explicit caps when attackers can inject larger-scale or structurally amplified malicious

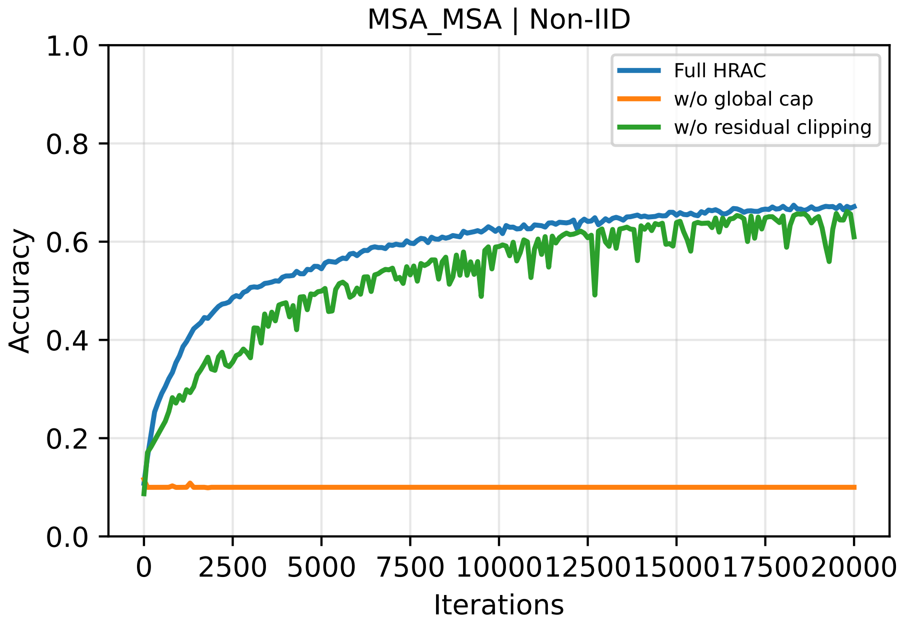

Figure 4.5: HRAC component ablation under MSA on non-IID CIFAR-10 with $b = 3$.

Table 4.5: Final and best test accuracy for the MSA non-IID ablation study.

Variant	**Final**	**Best**
Full HRAC	0.6713	0.6743
No global cap	0.1000	0.1163
No residual clipping	0.6099	0.6615

vectors.

4.3.3 Diagnostic Analysis

The HRAC logs also allow the accuracy results to be connected to the internal statistics used by the method. During training, the implementation records per-round residual thresholds τ_i, residual scale statistics μ_i, residual-change statistics ν_i, normalised client weights, pre- and post-clipping norms, and attacker-aware summaries in controlled settings where Byzantine identities are known. These identities are used only for evaluation and logging, not by the aggregation rule.

The final diagnostic plots are treated as supporting analysis rather than as the primary evidence for the method. They are intended to check whether elevated late-stage ν_i values correspond to persistent residual movement, whether the delayed weighting rule reduces the influence of clients with elevated residual-change statistics, and whether benign non-IID clients are systematically suppressed. The main empirical claims in this chapter therefore rest on the accuracy curves, final/best accuracy summaries, and component ablations, while diagnostics are used to interpret those results.

4.4 Discussion and Limitations

The experimental results support the main motivation for HRAC, but they also show why robust aggregation should be interpreted as a trade-off rather than a single ranking. HRAC is most useful when malicious behaviour appears as structured or persistent deviation from a client's historical path, as in MSA, HisMSA, non-IID ALIE, and non-IID IPM. In these regimes, comparing each client with its own history helps avoid the brittle assumption that all benign updates must cluster around one population centre.

The label-flipping and bit-flipping results are more mixed. Label flipping is a weaker and more protocol-compliant attack: clients train on corrupted labels but do not necessarily send geometrically extreme messages. Under strong heterogeneity, mean aggregation or FABA can therefore remain competitive. This does not invalidate HRAC, but it clarifies its role: HRAC is a robust aggregation strategy that reduces harmful influence softly; it is not a specialised label-poisoning detector.

The main limitation is hyperparameter sensitivity. The residual clipping multiplier c, the global cap multiplier c_g, and the ν penalty strength α control how aggressively HRAC suppresses client updates. Values that are too small can down-weight benign non-IID clients, while values that are too large may allow stealthy attacks to retain influence. The current defaults were kept fixed across the reported CIFAR-10 experiments, but a fuller study should include systematic sensitivity tests.

A second limitation is experimental coverage. The current results focus on CIFAR-10 with 10 clients and a fixed Byzantine level of $b = 3$. A stronger evaluation would vary the number of clients, attacker fraction, data heterogeneity, and model architecture, ideally across multiple random seeds.

5

Professional Issues and Project Management

Contents

5.1 Project Plan

The project was organised into work packages (WPs) covering literature review, baseline reproduction, attack integration, HRAC implementation, experimental evaluation, and final write-up. Early effort was concentrated on establishing a reliable experimental pipeline, including datasets, training loops, attack baselines, logging, and plotting, so that later conclusions would not be limited by tooling or reproducibility issues.

Progress was managed through weekly milestones defined by measurable outcomes rather than only dates. Core milestones included reproducing a stable benign baseline, validating each attack setting, integrating HRAC into the common training loop, and producing result tables, ablations, and diagnostic logs. When outcomes deviated from expectations, the plan was revised by updating estimates and re-prioritising tasks, particularly during phases where debugging and experimental variance dominated progress.

Risk management focused on training instability under non-IID partitions, hyperparameter sensitivity, and limited feedback cycles. Mitigations included staged evaluation from IID to non-IID settings, fixed configurations for main comparisons, and early production of intermediate results. A contingency path was also defined: if HRAC did not consistently outperform strong baselines, the project would still deliver a defensible contribution through ablation studies, failure-mode analysis, and clear reporting of limitations.

5.2 Evaluation Plan

The evaluation was structured around three questions: whether HRAC preserves benign utility, whether it improves robustness under Byzantine behaviour without oracle attacker labels, and whether its logged residual and weighting signals help explain the observed behaviour.

Results are reported across IID and non-IID partitions, with the main CIFAR-10 attack settings using $b = 3$ malicious clients out of $N = 10$. The attack set includes protocol-compliant label flipping, structural model manipulation, history-aware manipulation, and variance- or direction-based attacks. All methods are evaluated under matched training settings, and the limited seed and attacker-fraction coverage is reported explicitly in Chapter 4.

Success is measured primarily through test-accuracy trajectories, final and best accuracy, and component ablations. Diagnostic logs, including residual thresholds, μ/ν histories, normalised weights, and attacker-aware summaries, are used to support interpretation rather than to define success alone.

Component-level justification is provided through ablation studies that disable individual HRAC mechanisms while keeping the surrounding pipeline fixed. Reproducibility is supported by cached records, logged configurations, and standardised plots and tables across methods. Additional seeds, datasets, and attacker fractions remain important extensions beyond the current scope.

5.3 Legal and Ethical Matters

This project investigates robustness in federated/distributed learning under the Byzantine threat model. The work is conducted as a research and engineering study using established open-source software and publicly available benchmark datasets for academic evaluation. It does not involve deployment in a live operational environment or the collection of new personal data.

Legal considerations. The implementation relies on third-party libraries and datasets distributed under open licences. Their licence terms and attribution requirements are respected through appropriate citation and acknowledgement in the dissertation and accompanying codebase. From a data protection perspective, experiments are performed on standard public datasets rather than personal or sensitive user records. The project nevertheless recognises that, in real-world federated learning deployments, model updates may carry information that could be considered sensitive. Any future deployment would therefore require suitable governance and technical controls (e.g., access control, secure aggregation, and procedures consistent with applicable data protection regulation such as GDPR).

Ethical considerations. The project includes adversarial behaviours to evaluate robustness systematically; the intent is defensive and scientific rather than enabling misuse. Ethical risks are mitigated by focusing on aggregate results and interpretable diagnostic statistics, and by avoiding unnecessary dissemination of operational misuse guidance. The work also considers the ethical risk of unfairly penalising benign clients under non-IID data heterogeneity. To reduce such harm, HRAC compares clients mainly with their own histories and applies continuous suppression rather than rigid exclusion.

5.4 Impact

The main environmental impact of this project comes from compute used for repeated training runs (attacks, baselines, and ablations). This is limited by keeping experiments tightly scoped, reusing standard benchmarks, and logging results carefully to avoid unnecessary reruns. The defence itself is lightweight on the server side because it uses vector norm operations and per-client EMA states, so it does not add substantial overhead per round.

Societally, more robust federated learning can make privacy-preserving systems more reliable in areas such as edge sensing, healthcare, and industrial monitoring. The main risks are increased compute in large deployments and the possibility that benign but highly non-IID clients are down-weighted over time. These are reduced by using client-specific histories, soft weighting rather than hard exclusion, and participation diagnostics to catch unwanted behaviour early.

5.5 EDI

Equality, Diversity, and Inclusion considerations are reflected in the project's emphasis on heterogeneous client behaviour. Non-IID data is treated as a first-class condition, which in real deployments may correspond to diversity in users, devices, environments, and usage patterns. Robust aggregation methods can unintentionally disadvantage minority or atypical yet benign clients if they rely on tight clustering assumptions. HRAC mitigates this risk by using each client's historical residual behaviour as the main reference, aiming to remain tolerant to legitimate variability while still resisting adversarial manipulation.

In addition, the project maintains interpretable diagnostics (per-client residual statistics and weighting statistics) to support auditing and to identify systematic patterns of misclassification that could disproportionately affect particular client populations. These practices contribute to a more transparent and accountable engineering process for systems intended to operate across diverse participants.

5.6 Quality Management

Quality was maintained using standard software and experimental practice. All code and configurations were tracked with version control to support traceability and reproducibility. Experiments were run with consistent settings (including fixed random seeds where appropriate) and key metrics were logged in a structured way (e.g., accuracy, $\tau/\mu/\nu$ statistics, aggregation weights, and attacker-aware diagnostics).

Basic validation checks were built into the pipeline, including numerical stability safeguards and sanity checks on intermediate values (such as weight non-negativity and normalisation). We also ran benign-condition regression tests to ensure the defence does not degrade normal training, and used baseline comparisons and ablations to verify the effect of each component.

5.7 Safety Plan

This project is a software-only study on public benchmark datasets and does not involve laboratory equipment, hardware prototyping, human participants, or hazardous materials. Therefore, there are no significant physical safety risks.

The main remaining risks are minimal and relate to good computing practice: ensuring code and results are stored securely, avoiding any sensitive or personally identifiable data in datasets/logs, and documenting the experimental setup to prevent misuse or misinterpretation of results. Any future deployment in safety-critical domains would require a separate, domain-specific safety assessment.

Conclusions and future directions

This project investigated Byzantine robustness in federated/distributed learning, aiming to preserve utility under non-IID data while resisting both overt and stealthy attacks. The main outcome is HRAC, a history-residual adaptive clipping aggregator implemented in a modular framework with interchangeable attacks, aggregators, and diagnostic logging.

HRAC combines a median-based global norm cap with per-client residual clipping, so benign clients are compared primarily with their own histories rather than with a single global centre. Its behaviour is made interpretable through per-round residual thresholds, μ/ν history statistics, normalised client weights, and attacker-aware bottom-weight summaries in controlled settings. These diagnostics help explain robustness–utility trade-offs beyond final accuracy.

Empirically, HRAC remains competitive under benign CIFAR-10 training and shows stronger robustness in several tested structured or temporally persistent attack settings, particularly in non-IID MSA, HisMSA, ALIE, and IPM. The ablation study clarifies that different components matter in different regimes: the global median cap is essential under MSA, while ν-based temporal weighting is especially important under ALIE. These findings support the central design choice of combining round-wise magnitude control with client-specific residual histories.

Limitations include hyperparameter sensitivity, limited experimental coverage, and the possibility that highly non-IID but benign clients may still be down-weighted if their residual-change statistics remain elevated. The empirical results should therefore be read as controlled single-seed trends rather than as a complete statistical benchmark. Future work will (i) evaluate across multiple datasets, client populations, Byzantine fractions, and non-IID severities, (ii) compare against a broader set of baselines under matched training budgets, and (iii) improve self-calibration of thresholds and weights while testing stronger adaptive attackers that explicitly exploit client histories.

A
Convergence Analysis of HRAC

This appendix adapts the standalone convergence note in `prove.tex` to the thesis format. It contains the main proof chain needed for the report: HRAC is abstracted as a robust aggregation rule, the aggregate is compared with the ideal honest-client average, and the resulting aggregation error is inserted into a smooth non-convex descent argument. The proof has also been aligned with the implemented EMA-smoothed global norm cap used by the current HRAC code.

The result should be read as an optimisation-layer stability guarantee. It does not claim that arbitrary Byzantine behaviour disappears; instead, it shows that under explicit boundedness, clipping-distortion, and residual Byzantine-mass assumptions, HRAC converges to a neighbourhood of first-order stationarity. The size of that neighbourhood is controlled by honest stochastic variation, non-IID bias, clipping distortion, and the remaining Byzantine aggregation error.

A.1 Introduction

The goal of this document is to give a mathematically clean convergence analysis for HRAC. Rather than proving the code line by line, we abstract HRAC as a robust aggregation rule and proceed in two steps:

1. show that the HRAC aggregate remains close to the ideal honest average update;
2. show that such an aggregation error can be absorbed into the descent analysis of smooth non-convex optimisation.

For the adaptive statistics, the main theorem-level target is not universal pointwise ordering between Byzantine and honest clients at every late-stage round. Instead, the primary claim is the weaker and empirically better-matched one: Byzantine clients exhibit a larger late-stage group-mean ν-band, which induces stronger penalty pressure and smaller weights once the activation threshold becomes active. In particular, the intended mechanism is not that honest statistics vanish while Byzantine statistics remain nonzero. Rather, both groups may stay in positive late-stage bands, while Byzantine clients are typically elevated relative to the honest population. Pointwise separation is kept as a stronger sufficient refinement.

A.2 Problem Setup

Let $x_t \in \mathbb{R}^d$ denote the global model at round t. Suppose there are N participating clients, partitioned into an honest set $\mathcal{H}$ and a Byzantine set $\mathcal{B}$, with

$$|\mathcal{H}| = H, \qquad |\mathcal{B}| = B, \qquad N = H + B.$$

At round t, the server receives client updates

$$\{\Delta_i^t\}_{i=1}^N \subset \mathbb{R}^d,$$

and performs the global update

$$x_{t+1} = x_t - \eta_t g_t, \tag{A.1}$$

where g_t is the aggregate produced by HRAC.

The ideal honest average update is defined as

$$\bar{\Delta}_{\mathcal{H}}^t := \frac{1}{H} \sum_{i \in \mathcal{H}} \Delta_i^t. \tag{A.2}$$

The objective is to minimise a possibly non-convex function

$$f : \mathbb{R}^d \to \mathbb{R}.$$

A.3 HRAC Aggregation Rule

HRAC operates in three stages.

A.3.1 Global norm clipping

Define the round-wise median norm and its EMA-smoothed scale as

$$m_t := \mathrm{med}_{1 \leq j \leq N} \left\| \Delta_j^t \right\|, \tag{A.3}$$

$$\bar{m}_t := \rho_g \bar{m}_{t-1} + (1 - \rho_g) m_t, \tag{A.4}$$

with the first round warm-starting $\bar{m}$ from the observed median norm. The cap used during round t is the stored scale available at the start of the round,

$$B_t := c_g \bar{m}_{t-1}. \tag{A.5}$$

Thus the current round updates the next-round clipping scale, but does not immediately change its own threshold. The globally clipped update is

$$\hat{\Delta}_i^t := \mathrm{clip}(\Delta_i^t, B_t) := \begin{cases} \Delta_i^t, & \|\Delta_i^t\| \leq B_t, \\ B_t \dfrac{\Delta_i^t}{\|\Delta_i^t\|}, & \|\Delta_i^t\| > B_t. \end{cases} \tag{A.6}$$

A.3.2 Per-client history-residual clipping

For each client i, HRAC maintains a history state

$$(b_i^t, \mu_i^t, \nu_i^t),$$

where b_i^t is a running centre, μ_i^t is a scale estimate, and ν_i^t is a change statistic.

Define the residual

$$r_i^t := \hat{\Delta}_i^t - b_i^t, \tag{A.7}$$

and the adaptive residual clipping threshold

$$\tau_i^t := c\mu_i^t. \tag{A.8}$$

The clipped residual is

$$\bar{r}_i^t := \mathrm{clip}(r_i^t, \tau_i^t) = \begin{cases} r_i^t, & \|r_i^t\| \le \tau_i^t, \\ \tau_i^t \dfrac{r_i^t}{\|r_i^t\|}, & \|r_i^t\| > \tau_i^t. \end{cases} \tag{A.9}$$

The post-processed message becomes

$$\tilde{\Delta}_i^t := b_i^t + \bar{r}_i^t. \tag{A.10}$$

The history centre b_i^t follows the implemented EMA dynamics: for a newly appearing client, b_i^t is initialized by $\hat{\Delta}_i^t$; for later rounds,

$$b_i^{t+1} = \rho_b\, b_i^t + (1-\rho_b)\, \Delta_{i,\mathrm{safe}}^t, \tag{A.11}$$

where $\Delta_{i,\mathrm{safe}}^t$ is the aggregation-path message used in the server history update.

A.3.3 Adaptive weighting

HRAC assigns weights $w_i^t \ge 0$ satisfying

$$\sum_{i=1}^N w_i^t = 1, \tag{A.12}$$

where the weights are determined from the statistics $\{\nu_i^t\}_{i=1}^N$.

The final aggregate is

$$g_t = \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t. \tag{A.13}$$

A.4 Assumptions

Assumption 1 (Lower bounded objective)**.** *f is bounded below: $f(x) \ge f^\star$ for all $x \in \mathbb{R}^d$.*

Assumption 2 (L-smoothness). *$\|\nabla f(x) - \nabla f(y)\| \leq L\,\|x - y\|$ for all $x, y \in \mathbb{R}^d$.*

Assumption 3 (Bias and variance of the honest average). *Conditioned on $\mathcal{F}_t$, the honest average has bounded bias and variance:*

$$\left\|\mathbb{E}\left[\bar{\Delta}^t_{\mathcal{H}} \mid \mathcal{F}_t\right] - \nabla f(x_t)\right\| \leq \xi, \tag{A.14}$$

$$\mathbb{E}\left[\left\|\bar{\Delta}^t_{\mathcal{H}} - \mathbb{E}\left[\bar{\Delta}^t_{\mathcal{H}} \mid \mathcal{F}_t\right]\right\|^2 \,\middle|\, \mathcal{F}_t\right] \leq \frac{\sigma^2}{H}. \tag{A.15}$$

Assumption 4 (Bounded honest update magnitude). *There exists $G_H < \infty$ such that for all honest clients $i \in \mathcal{H}$ and all rounds t,*

$$\left\|\Delta^t_i\right\| \leq G_H. \tag{A.16}$$

Assumption 5 (Honest majority for robust median cap). *The Byzantine population is strictly less than the honest population:*

$$B < H \qquad (\textit{equivalently } B < N/2). \tag{A.17}$$

Assumption 6 (Uniform honest global-clipping distortion). *There exists a constant $\alpha_{\mathrm{gc}} \in [0, 1]$ such that for every honest client $i \in \mathcal{H}$ and every round t,*

$$\left\|\hat{\Delta}^t_i - \Delta^t_i\right\| \leq \alpha_{\mathrm{gc}} G_H. \tag{A.18}$$

Assumption 7 (Uniform honest residual-clipping slack). *There exists a constant $\alpha_{\mathrm{rc}} \in [0, 1]$ such that for every honest client $i \in \mathcal{H}$ and every round t,*

$$(\left\|r^t_i\right\| - \tau^t_i)_+ \leq \alpha_{\mathrm{rc}} G_H. \tag{A.19}$$

Assumption 8 (Uniform Byzantine mass and distortion control). *There exist constants $\beta_\infty \in [0, 1]$ and $C_w, C_m \geq 0$ such that for every round t,*

$$\beta_t \leq \beta_\infty, \tag{A.20}$$

$$\varepsilon_{w,t} \leq C_w\, \beta_t, \tag{A.21}$$

$$\sum_{j \in \mathcal{B}} w^t_j \left\|\tilde{\Delta}^t_j\right\|^2 \leq C_m\, \beta_t\, G_H^2. \tag{A.22}$$

Remark 1 (How to instantiate β_∞). Assumption 8 is stated abstractly so that the optimization theorem stays compatible with the implemented HRAC weighting rule. In an idealized exponential-weight model,

$$w_i^t = \frac{\exp(-\lambda \nu_i^t)}{\sum_{k=1}^N \exp(-\lambda \nu_k^t)},$$

if the late-stage separation condition $\nu_j^t \geq \nu_i^t + \gamma$ holds for all $j \in \mathcal{B}$, $i \in \mathcal{H}$, then one may take

$$\beta_\infty := \frac{Be^{-\lambda\gamma}}{H + Be^{-\lambda\gamma}}. \tag{A.23}$$

Thus the Byzantine contribution becomes a fully explicit closed constant.

A.5 Basic Properties of HRAC

Lemma 1 (Boundedness of globally clipped inputs)**.** *For every round t and every client i,*

$$\left\|\hat{\Delta}_i^t\right\| \leq B_t, \tag{A.24}$$

Proof. The bound $\left\|\hat{\Delta}_i^t\right\| \leq B_t$ follows directly from the definition of global clipping in (A.6).

□

Lemma 2 (EMA median-cap protection under honest majority)**.** *Under Assumptions 5 and 4, every round-wise median norm satisfies*

$$m_t = \mathrm{med}_{1\leq j\leq N} \left\|\Delta_j^t\right\| \leq \max_{i\in\mathcal{H}} \left\|\Delta_i^t\right\| \leq G_H. \tag{A.25}$$

If the EMA scale $\bar{m}_t$ is warm-started from such a median norm and updated by (A.4)*, then $\bar{m}_t \leq G_H$ for all rounds. Consequently,*

$$B_t = c_g \bar{m}_{t-1} \leq c_g G_H. \tag{A.26}$$

Proof. Because $H > B$, more than half of the submitted norms are honest. Hence the median order statistic cannot exceed the largest honest norm, giving (A.25). The EMA recursion for $\bar{m}_t$ is a convex combination of the previous scale and the current median norm. If both are at most G_H, the next scale is also at most G_H. Induction from the warm-start proves $\bar{m}_t \leq G_H$, and multiplying by c_g gives (A.26). □

Corollary 1 (Uniform cap bound from honest majority and bounded honest updates)**.** *Under Assumptions 5 and 4,*

$$B_t \leq c_g G_H =: B_{\max}, \qquad \forall t. \tag{A.27}$$

Proof. This is exactly the cap bound in Lemma 2. □

Lemma 3 (EMA boundedness for μ_i^t)**.** *Suppose nonnegative scalars s_t satisfy $s_t \le S_{\max}$ for all t, and $\mu_{t+1} = \rho_\mu \mu_t + (1-\rho_\mu)s_t$ with $0 < \rho_\mu < 1$ and $\mu_0 \le S_{\max}$. Then $\mu_t \le S_{\max}$ for all t.*

Proof. Induction with convexity:

$$\mu_{t+1} = \rho_\mu \mu_t + (1-\rho_\mu)s_t \le \rho_\mu S_{\max} + (1-\rho_\mu)S_{\max} = S_{\max}.$$

□

Lemma 4 (Bounded history states under bounded global cap)**.** *Assume there exists $B_{\max} < \infty$ such that $B_t \le B_{\max}$ for all rounds t, and the server-side history update uses the capped safe message ($\left\|\Delta_{i,\mathrm{safe}}^t\right\| \le B_t$, as in the implementation). Then for every client i and every t:*

$$\left\|b_i^t\right\| \le B_{\max}, \qquad \left\|r_i^t\right\| \le 2B_{\max}, \qquad \mu_i^t \le 2B_{\max}, \qquad \tau_i^t = c\mu_i^t \le 2cB_{\max}. \tag{A.28}$$

Moreover, the post-processed message is deterministically bounded:

$$\left\|\tilde{\Delta}_i^t\right\| \le (1+2c)B_{\max} := M_{\tilde{\Delta}}. \tag{A.29}$$

Proof. For b_i^t, the initialization satisfies $\left\|b_i^0\right\| \le B_{\max}$ since it is set from a globally clipped message. If $\|b_i^t\| \le B_{\max}$, then

$$b_i^{t+1} = \rho_b b_i^t + (1-\rho_b)\Delta_{i,\mathrm{safe}}^t, \qquad \left\|\Delta_{i,\mathrm{safe}}^t\right\| \le B_t \le B_{\max},$$

so convexity gives $\left\|b_i^{t+1}\right\| \le B_{\max}$. Hence $\|b_i^t\| \le B_{\max}$ for all t.

Because $\left\|\hat{\Delta}_i^t\right\| \le B_t \le B_{\max}$,

$$\left\|r_i^t\right\| = \left\|\hat{\Delta}_i^t - b_i^t\right\| \le \left\|\hat{\Delta}_i^t\right\| + \left\|b_i^t\right\| \le 2B_{\max}.$$

If μ_i^t is initialized at most $2B_{\max}$ and updated by EMA of bounded residual magnitudes (implementation path), then Lemma 3 gives $\mu_i^t \le 2B_{\max}$; thus $\tau_i^t = c\mu_i^t \le 2cB_{\max}$.

Finally, $\tilde{\Delta}_i^t = b_i^t + \bar{r}_i^t$ with $\|\bar{r}_i^t\| \le \tau_i^t$, so

$$\left\|\tilde{\Delta}_i^t\right\| \le \left\|b_i^t\right\| + \left\|\bar{r}_i^t\right\| \le B_{\max} + 2cB_{\max} = (1+2c)B_{\max}.$$

□

Lemma 5 (Aggregate second moment from deterministic cap chain)**.** *Under Assumptions 5 and 4, for every t, the aggregation output satisfies*

$$\mathbb{E}\Big[\|g_t\|^2 \mid \mathcal{F}_t\Big] \le M_{\tilde{\Delta}}^2, \qquad \textit{hence} \qquad \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\,\|g_t\|^2 \le M_{\tilde{\Delta}}^2. \tag{A.30}$$

Proof. By Corollary 1, $B_t \le B_{\max}$ for all t. Lemma 4 then gives $\left\|\tilde{\Delta}_i^t\right\| \le M_{\tilde{\Delta}}$ deterministically for every client i. Write $g_t = \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t$ with $\sum_i w_i^t = 1$, $w_i^t \ge 0$. Jensen's inequality yields:

$$\|g_t\|^2 = \left\|\sum_{i=1}^N w_i^t \tilde{\Delta}_i^t\right\|^2 \le \sum_{i=1}^N w_i^t \left\|\tilde{\Delta}_i^t\right\|^2 \le \sum_{i=1}^N w_i^t M_{\tilde{\Delta}}^2 = M_{\tilde{\Delta}}^2.$$

Taking expectations preserves the bound. □

Lemma 6 (Residual clipping bias identity)**.** *For every client i and round t,*

$$\|\bar{r}_i^t - r_i^t\| = (\|r_i^t\| - \tau_i^t)_+, \tag{A.31}$$

where $(a)_+ := \max\{a, 0\}$. As a consequence,

$$\left\|\tilde{\Delta}_i^t - \hat{\Delta}_i^t\right\| = (\|r_i^t\| - \tau_i^t)_+. \tag{A.32}$$

Proof. If $\|r_i^t\| \le \tau_i^t$, then $\bar{r}_i^t = r_i^t$, so the left-hand side is 0, equal to $(\|r_i^t\| - \tau_i^t)_+$.

If $\|r_i^t\| > \tau_i^t$, then

$$\bar{r}_i^t = r_i^t \cdot \frac{\tau_i^t}{\|r_i^t\|},$$

hence

$$\|\bar{r}_i^t - r_i^t\| = \left\|r_i^t\left(\frac{\tau_i^t}{\|r_i^t\|} - 1\right)\right\| = \|r_i^t\| - \tau_i^t.$$

This proves (A.31). Since

$$\tilde{\Delta}_i^t - \hat{\Delta}_i^t = (b_i^t + \bar{r}_i^t) - (b_i^t + r_i^t) = \bar{r}_i^t - r_i^t,$$

we obtain (A.32). □

A.6 Aggregation Error Analysis

Define the aggregation error

$$e_t := g_t - \bar{\Delta}^t_{\mathcal{H}}. \tag{A.33}$$

Lemma 7 (Aggregation error decomposition)**.** *The error e_t admits the decomposition*

$$e_t = \underbrace{\sum_{i\in\mathcal{H}} w_i^t(\tilde{\Delta}_i^t - \Delta_i^t)}_{=:A_t} + \underbrace{\sum_{i\in\mathcal{H}} \left(w_i^t - \frac{1}{H}\right)\Delta_i^t}_{=:W_t} + \underbrace{\sum_{j\in\mathcal{B}} w_j^t \tilde{\Delta}_j^t}_{=:M_t}. \tag{A.34}$$

Proof. From (A.13),

$$g_t = \sum_{i\in\mathcal{H}} w_i^t \tilde{\Delta}_i^t + \sum_{j\in\mathcal{B}} w_j^t \tilde{\Delta}_j^t.$$

Subtract $\bar{\Delta}^t_{\mathcal{H}} = \frac{1}{H}\sum_{i\in\mathcal{H}} \Delta_i^t$ and add/subtract $\sum_{i\in\mathcal{H}} w_i^t \Delta_i^t$ to obtain

$$g_t - \bar{\Delta}^t_{\mathcal{H}} = \sum_{i\in\mathcal{H}} w_i^t(\tilde{\Delta}_i^t - \Delta_i^t) + \sum_{i\in\mathcal{H}} \left(w_i^t - \frac{1}{H}\right)\Delta_i^t + \sum_{j\in\mathcal{B}} w_j^t \tilde{\Delta}_j^t.$$

□

Lemma 8 (Bounding each term of the decomposition)**.**

$$\|A_t\|^2 \le \sum_{i\in\mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2, \tag{A.35}$$

$$\|W_t\|^2 \le \varepsilon_{w,t}^2 \cdot \max_{i\in\mathcal{H}} \left\|\Delta_i^t\right\|^2, \tag{A.36}$$

$$\|M_t\|^2 \le \beta_t \sum_{j\in\mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|^2. \tag{A.37}$$

Consequently,

$$\|e_t\|^2 \le 3\|A_t\|^2 + 3\|W_t\|^2 + 3\|M_t\|^2. \tag{A.38}$$

Proof. For A_t, Jensen's inequality gives

$$\|A_t\|^2 = \left\|\sum_{i\in\mathcal{H}} w_i^t(\tilde{\Delta}_i^t - \Delta_i^t)\right\|^2 \le \sum_{i\in\mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2.$$

More explicitly, since $\sum_{i\in\mathcal{H}} w_i^t = 1 - \beta_t \le 1$, weighted Cauchy–Schwarz yields

$$\|A_t\|^2 \le \left(\sum_{i\in\mathcal{H}} w_i^t\right)\left(\sum_{i\in\mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2\right) \le \sum_{i\in\mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2.$$

For W_t,

$$\|W_t\| \le \sum_{i\in\mathcal{H}} \left| w_i^t - \frac{1}{H} \right| \left\|\Delta_i^t\right\| \le \varepsilon_{w,t} \max_{i\in\mathcal{H}} \left\|\Delta_i^t\right\|,$$

which yields (A.36) after squaring.

For M_t, by Cauchy–Schwarz and $\sum_{j\in\mathcal{B}} w_j^t = \beta_t$,

$$\|M_t\| \le \sum_{j\in\mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|$$

and therefore

$$\|M_t\|^2 \le \left(\sum_{j\in\mathcal{B}} w_j^t\right) \left(\sum_{j\in\mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|^2\right) = \beta_t \sum_{j\in\mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|^2.$$

which is exactly (A.37).

Finally,

$$e_t = A_t + W_t + M_t,$$

so

$$\|e_t\|^2 \le 3\|A_t\|^2 + 3\|W_t\|^2 + 3\|M_t\|^2.$$

□

Lemma 9 (Explicit honest clipping bias bound)**.** *For every honest client $i \in \mathcal{H}$,*

$$\left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2 \le 2\left\|\hat{\Delta}_i^t - \Delta_i^t\right\|^2 + 2(\|r_i^t\| - \tau_i^t)_+^2. \tag{A.39}$$

Therefore,

$$\sum_{i\in\mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2 \le 2\sum_{i\in\mathcal{H}} w_i^t \left\|\hat{\Delta}_i^t - \Delta_i^t\right\|^2 + 2\sum_{i\in\mathcal{H}} w_i^t(\|r_i^t\| - \tau_i^t)_+^2. \tag{A.40}$$

Proof. Using

$$\tilde{\Delta}_i^t - \Delta_i^t = (\tilde{\Delta}_i^t - \hat{\Delta}_i^t) + (\hat{\Delta}_i^t - \Delta_i^t),$$

we have

$$\left\|\tilde{\Delta}_i^t - \Delta_i^t\right\|^2 \le 2\left\|\tilde{\Delta}_i^t - \hat{\Delta}_i^t\right\|^2 + 2\left\|\hat{\Delta}_i^t - \Delta_i^t\right\|^2.$$

Lemma 6 gives

$$\left\|\tilde{\Delta}_i^t - \hat{\Delta}_i^t\right\|^2 = (\|r_i^t\| - \tau_i^t)_+^2.$$

Substituting yields (A.39). Summing with weights w_i^t gives (A.40). □

Lemma 10 (Overall aggregation error bound)**.**

$$\|e_t\|^2 \le 6\sum_{i\in\mathcal{H}} w_i^t \left\|\hat{\Delta}_i^t - \Delta_i^t\right\|^2 + 6\sum_{i\in\mathcal{H}} w_i^t(\|r_i^t\| - \tau_i^t)_+^2$$
$$+ 3\varepsilon_{w,t}^2 \max_{i\in\mathcal{H}} \left\|\Delta_i^t\right\|^2 + 3\beta_t \sum_{j\in\mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|^2 . \tag{A.41}$$

Proof. Combine Lemma 8 with Lemma 9. □

A.7 From Honest Average to the True Gradient

Lemma 11 (Deviation of the honest average from the true gradient)**.** *Under Assumption 3,*

$$\mathbb{E}\left[\left\|\bar{\Delta}_{\mathcal{H}}^t - \nabla f(x_t)\right\|^2 \,\middle|\, \mathcal{F}_t\right] \le \frac{\sigma^2}{H} + \xi^2. \tag{A.42}$$

Proof. Write

$$\bar{\Delta}_{\mathcal{H}}^t - \nabla f(x_t) = \left(\bar{\Delta}_{\mathcal{H}}^t - \mathbb{E}[\bar{\Delta}_{\mathcal{H}}^t \mid \mathcal{F}_t]\right) + \left(\mathbb{E}[\bar{\Delta}_{\mathcal{H}}^t \mid \mathcal{F}_t] - \nabla f(x_t)\right).$$

Using the variance bound (A.15) and the bias bound (A.14) gives the result. □

A.8 One-Step Descent Analysis

Lemma 12 (One-step descent inequality)**.** *Suppose Assumptions 1 to 5 hold. Then, for any step size $\eta_t > 0$,*

$$\mathbb{E}[f(x_{t+1})] \le \mathbb{E}[f(x_t)] - \frac{\eta_t}{2}\mathbb{E}\|\nabla f(x_t)\|^2$$
$$+ \eta_t \mathbb{E}\|e_t\|^2 + \eta_t\left(\frac{\sigma^2}{H} + \xi^2\right) + \frac{L\eta_t^2}{2}\mathbb{E}\|g_t\|^2 . \tag{A.43}$$

Proof. By Assumption 2,

$$f(x_{t+1}) \le f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t\rangle + \frac{L}{2}\|x_{t+1} - x_t\|^2 .$$

Substituting $x_{t+1} = x_t - \eta_t g_t$ gives

$$f(x_{t+1}) \le f(x_t) - \eta_t\langle \nabla f(x_t), g_t\rangle + \frac{L\eta_t^2}{2}\|g_t\|^2 .$$

Using the identity $-\langle a, b\rangle = -\frac{1}{2}\|a\|^2 - \frac{1}{2}\|b\|^2 + \frac{1}{2}\|a-b\|^2$, we can rewrite the inner product exactly as:

$$-\langle \nabla f(x_t), g_t\rangle = -\frac{1}{2}\|\nabla f(x_t)\|^2 - \frac{1}{2}\|g_t\|^2 + \frac{1}{2}\|g_t - \nabla f(x_t)\|^2 .$$

Since $-\frac{1}{2}\|g_t\|^2 \leq 0$, we drop it for an upper bound:

$$-\langle \nabla f(x_t), g_t\rangle \leq -\frac{1}{2}\|\nabla f(x_t)\|^2 + \frac{1}{2}\|g_t - \nabla f(x_t)\|^2 .$$

Now, using the decomposition $g_t = \bar{\Delta}^t_{\mathcal{H}} + e_t$, and applying the inequality $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, we bound the distance between the aggregate and the true gradient:

$$\|g_t - \nabla f(x_t)\|^2 = \left\|e_t + \bar{\Delta}^t_{\mathcal{H}} - \nabla f(x_t)\right\|^2 \leq 2\|e_t\|^2 + 2\left\|\bar{\Delta}^t_{\mathcal{H}} - \nabla f(x_t)\right\|^2 .$$

Substituting this back into the smoothness inequality yields:

$$f(x_{t+1}) \leq f(x_t) - \frac{\eta_t}{2}\|\nabla f(x_t)\|^2 + \eta_t\|e_t\|^2 + \eta_t\left\|\bar{\Delta}^t_{\mathcal{H}} - \nabla f(x_t)\right\|^2 + \frac{L\eta_t^2}{2}\|g_t\|^2 .$$

Taking the conditional expectation $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ on both sides and applying Lemma 11 for the honest average deviation yields the final result (A.43). □

A.9 Main Convergence Theorem

Proposition 1 (Uniform constant aggregation-error bound)**.** *Under Assumptions 4, 6, 7, and 8, the HRAC aggregation error satisfies, for every round t,*

$$\mathbb{E}\|e_t\|^2 \leq C_{\text{agg}}, \tag{A.44}$$

where

$$C_{\text{clip}} := 6\alpha_{\text{gc}}^2 G_H^2, \tag{A.45}$$

$$C_{\text{res}} := 6\alpha_{\text{rc}}^2 G_H^2, \tag{A.46}$$

$$C_{\text{byz}} := 3(C_w^2 + C_m)\beta_\infty^2 G_H^2, \tag{A.47}$$

$$C_{\text{agg}} := C_{\text{clip}} + C_{\text{res}} + C_{\text{byz}}. \tag{A.48}$$

Proof. By Assumption 6,

$$6 \sum_{i\in\mathcal{H}} w_i^t \left\| \hat{\Delta}_i^t - \Delta_i^t \right\|^2 \leq 6 \sum_{i\in\mathcal{H}} w_i^t \alpha_{\mathrm{gc}}^2 G_H^2 \leq 6\alpha_{\mathrm{gc}}^2 G_H^2 = C_{\mathrm{clip}}.$$

By Assumption 7,

$$6 \sum_{i\in\mathcal{H}} w_i^t (\|r_i^t\| - \tau_i^t)_+^2 \leq 6 \sum_{i\in\mathcal{H}} w_i^t \alpha_{\mathrm{rc}}^2 G_H^2 \leq 6\alpha_{\mathrm{rc}}^2 G_H^2 = C_{\mathrm{res}}.$$

By Assumptions 4 and 8,

$$3\varepsilon_{w,t}^2 \max_{i\in\mathcal{H}} \left\|\Delta_i^t\right\|^2 \leq 3(C_w\beta_t)^2 G_H^2 \leq 3C_w^2\beta_\infty^2 G_H^2,$$

and

$$3\beta_t \sum_{j\in\mathcal{B}} w_j^t \left\| \tilde{\Delta}_j^t \right\|^2 \leq 3\beta_t (C_m \beta_t G_H^2) \leq 3C_m \beta_\infty^2 G_H^2.$$

Combining these two displays yields C_{byz}. Finally, apply Lemma 10. □

Theorem 1 (Convergence to a tight HRAC constant neighbourhood)**.** *Suppose Assumptions 1 to 8 hold. Let the step size be constant and satisfy*

$$0 < \eta \leq \frac{1}{2L}. \tag{A.49}$$

Define

$$G_g := M_{\tilde{\Delta}}. \tag{A.50}$$

Define the explicit HRAC neighbourhood radius

$$R_{\mathrm{HRAC}} := 2\left(\frac{\sigma^2}{H} + \xi^2\right) + L\eta G_g^2 + 2C_{\mathrm{agg}}. \tag{A.51}$$

Then, for every $T \geq 1$,

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left\|\nabla f(x_t)\right\|^2 \leq \frac{2(f(x_0) - f^\star)}{\eta T} + R_{\mathrm{HRAC}}. \tag{A.52}$$

Consequently,

$$\limsup_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left\|\nabla f(x_t)\right\|^2 \leq R_{\mathrm{HRAC}}. \tag{A.53}$$

Therefore HRAC converges to a first-order stationary neighbourhood with an explicit, time-independent *radius.*

Proof. Sum the one-step descent inequality (A.43) from $t = 0$ to $T-1$:

$$\mathbb{E}[f(x_T)] \leq f(x_0) - \frac{\eta}{2}\sum_{t=0}^{T-1}\mathbb{E}\,\|\nabla f(x_t)\|^2 + \eta\sum_{t=0}^{T-1}\mathbb{E}\,\|e_t\|^2 + \eta T\left(\frac{\sigma^2}{H} + \xi^2\right) + \frac{L\eta^2}{2}\sum_{t=0}^{T-1}\mathbb{E}\,\|g_t\|^2 .$$

Since $f(x_T) \geq f^\star$, rearranging gives

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\,\|\nabla f(x_t)\|^2 \leq \frac{2(f(x_0) - f^\star)}{\eta T} + \frac{2}{T}\sum_{t=0}^{T-1}\mathbb{E}\,\|e_t\|^2 + 2\left(\frac{\sigma^2}{H} + \xi^2\right) + \frac{L\eta}{T}\sum_{t=0}^{T-1}\mathbb{E}\,\|g_t\|^2 .$$

By Lemma 5, $\mathbb{E}\,\|g_t\|^2 \leq M_{\tilde{\Delta}}^2$ for every t, and by Proposition 1, $\mathbb{E}\,\|e_t\|^2 \leq C_{\text{agg}}$ for every t. Substituting these uniform bounds yields (A.52). Taking $\limsup$ eliminates the vanishing optimization term $2(f(x_0) - f^\star)/(\eta T)$ and gives (A.53). □

Corollary 2 (Closed-form HRAC constant interval)**.** *Under the assumptions of Theorem 1,*

$$R_{\text{HRAC}} = 2\left(\frac{\sigma^2}{H} + \xi^2\right) + L\eta G_g^2 + 12\alpha_{\text{gc}}^2 G_H^2 + 12\alpha_{\text{rc}}^2 G_H^2 + 6(C_w^2 + C_m)\beta_\infty^2 G_H^2. \tag{A.54}$$

Hence the averaged stationarity measure is asymptotically confined to the explicit interval

$$0 \leq \limsup_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\,\|\nabla f(x_t)\|^2 \leq R_{\text{HRAC}}. \tag{A.55}$$

Remark 2 (Same theorem in schematic constant form). If one prefers the theorem-level presentation

$$R_{\text{HRAC}} = 2\left(\frac{\sigma^2}{H} + \xi^2\right) + L\eta G_g^2 + C_{\text{clip}}^{\text{eff}} + C_{\text{res}}^{\text{eff}} + C_{\text{byz}}^{\text{eff}},$$

then the effective constants in the present proof are

$$C_{\text{clip}}^{\text{eff}} := 12\alpha_{\text{gc}}^2 G_H^2, \qquad C_{\text{res}}^{\text{eff}} := 12\alpha_{\text{rc}}^2 G_H^2, \qquad C_{\text{byz}}^{\text{eff}} := 6(C_w^2 + C_m)\beta_\infty^2 G_H^2.$$

The factor 2 relative to the raw aggregation-error constants (A.45)–(A.47) comes from the descent step $\frac{2}{T}\sum_t \mathbb{E}\,\|e_t\|^2$ in (A.52).

Corollary 3 (Interpretation as a tight stationary neighbourhood)**.** *Under the assumptions of*

Theorem 1, HRAC converges sublinearly to a first-order stationary neighbourhood of radius R_{HRAC}*. In particular, the long-run averaged squared gradient norm is controlled by a tight constant composed of:*

1. *stochastic honest estimation error* $2(\sigma^2/H + \xi^2)$*;*
2. *optimization discretization error* $L\eta M_{\tilde{\Delta}}^2$*;*
3. *honest global-clipping distortion* $12\alpha_{\mathrm{gc}}^2 G_H^2$*;*
4. *honest residual-clipping distortion* $12\alpha_{\mathrm{rc}}^2 G_H^2$*;*
5. *residual Byzantine mass and honest-weight distortion* $6(C_w^2 + C_m)\beta_\infty^2 G_H^2$.

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